

Emmalie Cole

Cedar Park, TX • emmalie.cole@gmail.com • (360) 926-6661 • [linkedin.com/in/es-cole](https://www.linkedin.com/in/es-cole) • knead.it.com • i-ekoul.github.io

Senior Software Engineer — Agentic AI Systems & Full-Stack Product

AI-native full-stack engineer building production web, mobile, and AI systems end-to-end. Designs, operates, and evaluates multi-agent systems as everyday development infrastructure, working daily in Claude Code and Cursor. Founder of [knead.it](https://knead.it.com), a cross-platform culinary platform engineered solo through a 16-agent AI organization (Next.js + Capacitor, Neon Postgres, Claude API); at Endgame AI, leads an AI-assisted engineering team building a real-time chess platform across Flutter mobile, Vue 3 web, and event-driven C#/.NET + Python microservices on AWS.

TECHNICAL SKILLS

Agentic AI Systems: multi-agent orchestration & evaluation | agent eval sets & regression gates | voice agents (OpenAI TTS, voice-first UX) | MCP | Anthropic Claude API | Claude Code & Cursor (daily drivers) | LLM product engineering

Languages: Python | TypeScript | Dart | C# | JavaScript | SQL | HTML/CSS

Backend & Cloud: FastAPI | Pydantic (typed schemas & validation) | ASP.NET Core | SignalR (WebSockets) | Kafka | Node.js | MongoDB (change streams) | Neon Postgres | REST APIs | AWS (ECS, Lambda, ECR) | GCP (Cloud Run, BigQuery, Firestore) | Cloudflare Workers

Frontend & Mobile: Flutter | Vue 3 | React | Next.js | Capacitor | Vite | TailwindCSS

DevOps & Infrastructure: Docker | GitHub Actions | Terraform | CI/CD pipelines | distributed event-driven microservices | observability & crash monitoring | Playwright + Appium automation

Practices: system design | API & SDK design | developer experience | design & code reviews | ADRs | TDD | unit/integration/E2E coverage | Agile/Scrum | cross-functional collaboration

PROFESSIONAL EXPERIENCE

Founder & Principal Engineer — [knead.it](https://knead.it.com)

2025 – PRESENT

Cross-platform culinary platform: recipe library, interactive culinary globe, place-based cooking lessons | knead.it.com

- Conceived and architected an entire consumer platform solo, from product vision, brand, and UX through data model and API design, targeting web, iOS, and Android from one **Next.js + Capacitor** codebase on **Neon serverless Postgres**, **Cloudflare Workers**, and **Cloudflare Pages**
- Designed and operate an **agentic AI development organization**: 16 specialized Claude agents (backend, QA, brand, legal, research...) with scoped charters and plugin-style ownership boundaries, coordinated through 33 architecture decision records and cross-agent handoff protocols
- Built the agents' **evaluation and regression infrastructure**: AI-tutor eval set, automated brand linter, provenance gates for factual claims, and **Playwright + axe-core** E2E/accessibility/device-matrix suites gating every merge in **GitHub Actions** CI
- Built an AI cooking tutor on the **Anthropic Claude API** with hands-free voice guidance via **OpenAI TTS**, deployed as serverless Cloudflare Workers with origin-allowlist access control

Software Engineer & Product Manager — [Endgame AI](https://endgame.ai)

JAN 2026 – PRESENT

Early-stage chess-technology startup | Remote

- Led the engineering team, reporting directly to the CEO and COO: owned sprint planning, delivery cadence, and engineer onboarding/offboarding across a distributed org
- Set UX/UI design direction for the platform's web and mobile surfaces
- Build and ship across the full platform: **Flutter/Dart** mobile (iOS + Android), **Vue 3/TypeScript** web client, and **C#/.NET + Python** backend services in a 40+ microservice architecture
- Engineer live tournament broadcasting: **ASP.NET Core SignalR** WebSocket APIs backed by **MongoDB change streams**, fed by a multi-source, **Pydantic**-validated Python ingestion pipeline (Lichess API, DGT boards) on **AWS ECS + Lambda**
- Develop event-driven backend services on **Kafka** and the core game API (.NET, MongoDB), including shared developer libraries (Kafka producer/consumer, chess move-generation SDK) distributed as internal packages and deployed via **GitHub Actions** CI/CD to AWS ECR
- Own the mobile release cycle end-to-end (automated TestFlight/Play Store pipelines, **Playwright + Appium** UI automation, crash monitoring) and drive product strategy as PM across a distributed team
- Created the company's operations dashboard solo (**Next.js 14** on **GCP Cloud Run** behind IAP), used daily by the team and its investors: role-based access, analytics unifying **GA4**, **MongoDB**, **Firebase**, and **BigQuery**, z-score anomaly alerting
- Administer company billing and vendor accounts across the engineering toolchain

Full-Stack Software Engineer — [SmallChess](https://smallchess.com)

MAR – OCT 2025

Chess-technology startup | Educational & analytical tools | Remote

- Developed full-stack features for a chess training platform using **React**, **TypeScript**, **Supabase**, and **Docker**
- Designed backend services and database schema powering dynamic lesson delivery, PGN parsing, and Stockfish-integrated analysis engine
- Built move-prediction logic and adaptive lesson-progression system, improving the platform's ability to personalize learning paths per student skill level

EDUCATION: BS, Computer Science (Software Engineering), Southern New Hampshire University